

VOLUME 4

Wiring Diagrams and I/O Reference

Mazak VQC-20/40 SN 060231 - LinuxCNC/Mesa Retrofit

REVISION B - CONSOLIDATED FROM LIVE AUTHORITY

Field	Value
Document	MZK-LCNC-004
Revision	B
Generated	2026-08-07
Source commit	3e9c4e6
Machine	Mazak VQC-20/40, serial 060231
Original control	Mazatrol M-2 / MELDAS
Target control	LinuxCNC 2.9.10, Debian 13 PREEMPT_RT

STATUS - NOT READY FOR POWER OR MOTION

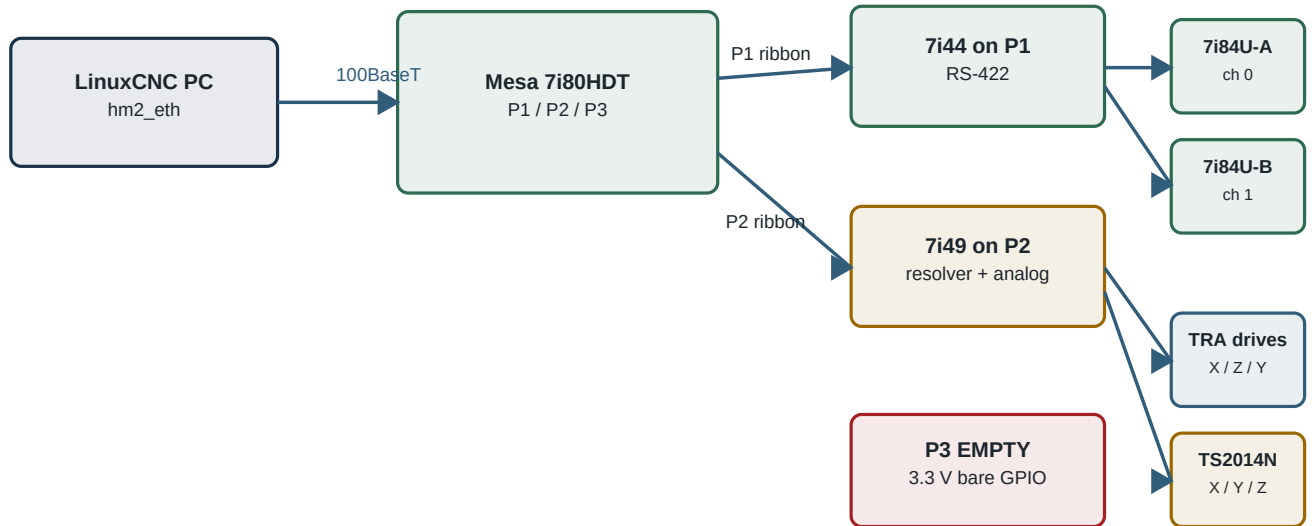
The repository is still at planning/bring-up stage. No live-power hold point is signed. Authority rows marked PROPOSED, COMMISSIONING_PENDING, HOLD_CONFLICT, or UNBOUND are not as-built evidence. The hardwired E-stop chain, resolver suffixes/scales, bitfile, analog polarity/scaling, field normal states, shared-bus behavior, and spindle terminals remain to be proven.

This volume replaces Rev A. Rev A must not be used for wiring, commissioning, or service.

WIRING AUTHORITY STATUS

Every table below is generated from mesa/current_pin_authority.csv. A row is not permission to land or energize a conductor unless its evidence state and applicable hold point allow it. HOLD_CONFLICT and UNBOUND rows are prohibited from field connection or actuation.

1. System architecture



Only the 7i49 carries motion feedback and analog commands. Field I/O is isolated through two 7i84Us. P3 has no field wiring.

Figure 4-1. Physical card topology and daughter-card ownership.

2. Servo command, enable, and feedback

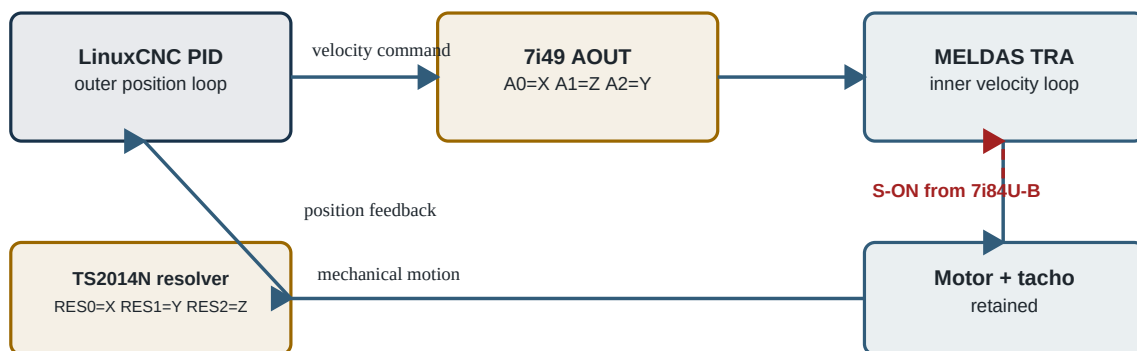
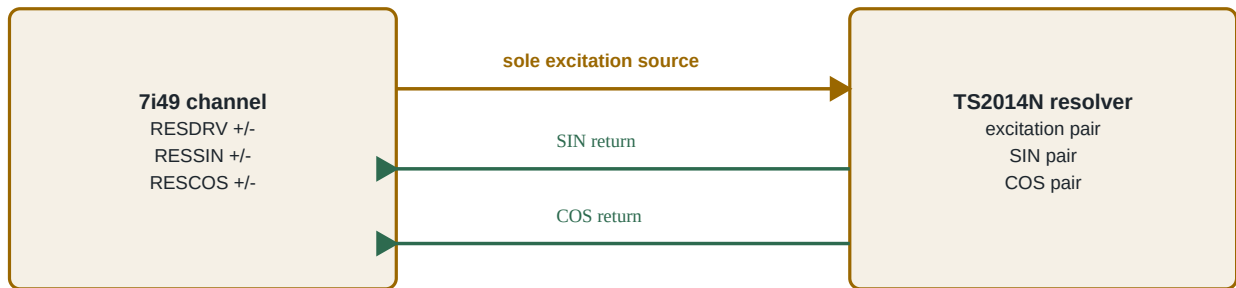


Figure 4-2. Velocity-mode nested loops and distinct 7i49/7i84U paths.

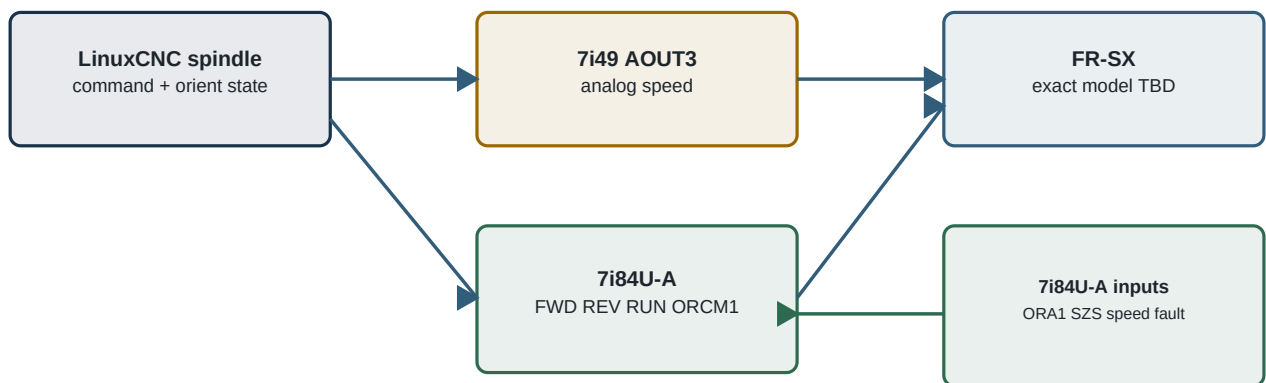
3. Resolver cable topology



Three individually shielded twisted pairs per axis; shields terminate at the 7i49 end only. Identify windings by resistance before power.

Figure 4-3. One 7i49 resolver channel; exact winding pairs are measured, not assumed.

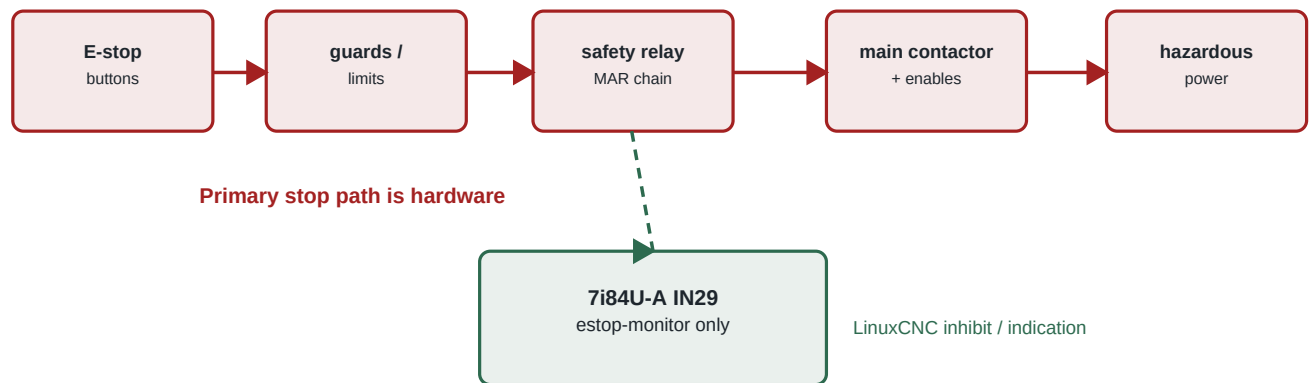
4. FR-SX spindle interface



Orient position is not an analog reference. ORCM1 is a discrete command; the machine sensor and drive parameters define the target.

Figure 4-4. Analog speed plus discrete run/orient model; exact FR-SX terminals remain unverified.

5. E-stop authority split



The as-built hardware chain and fault-injection matrix are incomplete. No SIL/PL or stop-category claim is made.

Figure 4-5. Hardware-primary stop path and software monitor tap.

6. 7i49 resolver and analog authority

Channel	Signal ID	HAL net	Field point / load	Status
RES0	X_RESOLVER	x-pos-fb	Tamagawa TS2014N X resolver	COMMISSIONING_PENDING
RES1	Y_RESOLVER	y-pos-fb	Tamagawa TS2014N Y resolver	COMMISSIONING_PENDING
RES2	Z_RESOLVER	z-pos-fb	Tamagawa TS2014N Z resolver	COMMISSIONING_PENDING
RES3	RES3_SPARE	none	7i49 spare resolver channel	SPARE
RES4	RES4_SPARE	none	7i49 spare resolver channel	SPARE
RES5	RES5_SPARE	none	7i49 spare resolver channel	SPARE
AOUT0	X_AXIS_CMD	x-vel-cmd	X servo analog velocity command to MELDAS TRA	COMMISSIONING_PENDING
AOUT1	Z_AXIS_CMD	z-vel-cmd	Z servo analog velocity command to MELDAS TRA	COMMISSIONING_PENDING
AOUT2	Y_AXIS_CMD	y-vel-cmd	Y servo analog velocity command to MELDAS TRA	COMMISSIONING_PENDING
AOUT3	SPINDLE_SPEED_CMD	spindle-speed-cmd	FR-SX V-IN speed reference	COMMISSIONING_PENDING
AOUT4	AOUT4_SPARE	none	Spare 7i49 analog output	SPARE
AOUT5	AOUT5_SPARE	none	Spare 7i49 analog output	SPARE

Axis analog order is X=AOUT0, Z=AOUT1, Y=AOUT2. Resolver order is X=RES0, Y=RES1, Z=RES2. These two orderings are deliberately different.

7. 7i44 smart-serial link authority

Channel	Signal ID	HAL net	Field point / load	Status
port0.TX+	SSERIAL_PORT0_TXA	none	7i84U-A RJ45 pin 2 RX+	PROPOSED
port0.TX-	SSERIAL_PORT0_TXB	none	7i84U-A RJ45 pin 1 RX-	PROPOSED
port0.RX+	SSERIAL_PORT0_RXA	none	7i84U-A RJ45 pin 6 TX+	PROPOSED
port0.RX-	SSERIAL_PORT0_RXB	none	7i84U-A RJ45 pin 3 TX-	PROPOSED
port0.GND	SSERIAL_PORT0_GND	none	7i84U-A RJ45 pin 4/5 ground	PROPOSED
port0.+5V	SSERIAL_PORT0_5V	none	7i84U-A RJ45 pin 7/8 +5V	PROPOSED
ports2-7	SSERIAL_PORTS_2_7_SPARE	none	Spare smart-serial channels	SPARE
port1.TX+	SSERIAL_PORT1_TXA	none	7i84U-B RJ45 pin 2 RX+	PROPOSED
port1.TX-	SSERIAL_PORT1_TXB	none	7i84U-B RJ45 pin 1 RX-	PROPOSED
port1.RX+	SSERIAL_PORT1_RXA	none	7i84U-B RJ45 pin 6 TX+	PROPOSED
port1.RX-	SSERIAL_PORT1_RXB	none	7i84U-B RJ45 pin 3 TX-	PROPOSED
port1.GND	SSERIAL_PORT1_GND	none	7i84U-B RJ45 pin 4/5 ground	PROPOSED
port1.+5V	SSERIAL_PORT1_5V	none	7i84U-B RJ45 pin 7/8 +5V	PROPOSED

8. 7i84U-A TB1 power map

Channel	Signal ID	HAL net	Field point / load	Status
TB1 pins 3/4 VFIELDA (+24 V field power, 5-28 VDC)	SEVENI84U_FIELD_A_24V	none	Field power bank A (TB3 I/O)	COMMISSIONING_PENDING
TB1 pins 1/2 VFIELDB (+24 V field power, 5-28 VDC)	SEVENI84U_FIELD_B_24V	none	Field power bank B (TB2 I/O)	COMMISSIONING_PENDING
TB1 pin 5 VIN (+24 V logic power)	SEVENI84U_VIN_24V	none	Field I/O logic power	COMMISSIONING_PENDING
TB1 pins 6/7/8 GND (VIN/VFIELD common)	SEVENI84U_GND	none	VIN/VFIELD common return	COMMISSIONING_PENDING

A-CARD TB1

Pins 1/2 are both VFIELDB positive; pins 3/4 are both VFIELDA positive; pin 5 is VIN; pins 6/7/8 are common. Use 5-28 VDC and record W1.

9. 7i84U-A inputs IN0-IN31

Channel	Signal ID	HAL net	Field point / load	Status
IN0	ATC_ZONE_Y	atc-y-zone	PRS-55 Y ATC zone	COMMISSIONING_PENDING
IN1	ATC_ZONE_Z	atc-z-zone	PRS-66 Z ATC zone	COMMISSIONING_PENDING
IN2	MAG_TOOL_AVAILABLE	mag-tool-avail	PHS-181 magazine tool available	COMMISSIONING_PENDING
IN3	SPINDLE_TOOL_AVAILABLE	spindle-tool-avail	PHS-182 spindle tool available	COMMISSIONING_PENDING
IN4	SPINDLE_ORIENT_ARRIVAL	spindle-oriented	FR-SX orient arrival (PLC X003 ORA1)	PROPOSED
IN5	SPINDLE_ZERO_SPEED	spindle-zero-speed	FR-SX zero-speed output (PLC X001 SZS.M)	PROPOSED
IN6	MAG_COVER_OPEN_CONF	mag-cover-open-conf	Magazine cover open reed switch (PLC X052 MGCORS)	PROPOSED
IN7	MAG_COVER_CLOSE_CONF	mag-cover-closed-conf	Magazine cover close reed switch (PLC X053 MGCCRS)	PROPOSED
IN8	THERMAL_ALARM_CHAIN	thermal-alarm	Motor thermal trip + main transformer overheat in series (PLC X073 THR.M + X07B ONT.M)	PROPOSED
IN9	MANUAL_TOOL_UNCLAMP_PB	manual-unclamp-pb	Manual tool unclamp switch at head (PLC X01A TUCFS.M)	PROPOSED
IN10	X_DRIVE_FAULT	x-drive-fault	X drive fault relay	COMMISSIONING_PENDING
IN11	Y_DRIVE_FAULT	y-drive-fault	Y drive fault relay	COMMISSIONING_PENDING
IN12	Z_DRIVE_FAULT	z-drive-fault	Z drive fault relay	COMMISSIONING_PENDING
IN13	SPINDLE_AT_SPEED	spindle-at-speed	FR-SX speed reach output	COMMISSIONING_PENDING
IN14	SPINDLE_FAULT	spindle-fault	FR-SX fault output	COMMISSIONING_PENDING
IN15	TOOL_CLAMP_CONF	tool-clamped	PRS-9 tool clamp confirm	COMMISSIONING_PENDING
IN16	TOOL_UNCLAMP_CONF	tool-unclamped	PRS-8 tool unclamp confirm	COMMISSIONING_PENDING
IN17	GEAR_HI_CONF	gear-hi-conf	PRS-10 gear high confirm	COMMISSIONING_PENDING
IN18	GEAR_LO_CONF	gear-lo-conf	PRS-12 gear low confirm	COMMISSIONING_PENDING
IN19	MAG_BCD_BIT0	mag-bcd-bit0	PRS-21 magazine BCD bit 0	COMMISSIONING_PENDING
IN20	MAG_BCD_BIT1	mag-bcd-bit1	PRS-22 magazine BCD bit 1	COMMISSIONING_PENDING
IN21	MAG_BCD_BIT2	mag-bcd-bit2	PRS-23 magazine BCD bit 2	COMMISSIONING_PENDING
IN22	MAG_BCD_BIT3	mag-bcd-bit3	PRS-24 magazine BCD bit 3	COMMISSIONING_PENDING
IN23	MAG_BCD_BIT4	mag-bcd-bit4	PRS-25 magazine BCD bit 4	COMMISSIONING_PENDING
IN24	DOOR_INTERLOCK	door-interlock	Door interlock switches series (LS-141 + LS-140 + PLC X01D ITMDSS.M all in one chain)	COMMISSIONING_PENDING
IN25	LUBE_OK	lube-ok	Two head-lube pressure switches (PLC X042 HLP2.M + X079 HLP.M) series-wired NC	PROPOSED
IN26	COOLANT_LOW	coolant-low	Coolant level switch	COMMISSIONING_PENDING
IN27	HYD_PRESS_OK	hydraulic-ok	Sanwa SPS-8T-PC-20 pressure switch	COMMISSIONING_PENDING
IN28	MAG_IN_POS	mag-in-pos	Magazine index in-position prox (PLC X00D MIPRS); BCD pot number valid only while TRUE	PROPOSED
IN29	ESTOP_MONITOR	estop-monitor	OEM MAR relay aux contact via interposing relay (Omron G2R-1-SND-DC24 or Phoenix PLC-RSC-24DC/21) driven from EHB bus	PROPOSED
IN30	MANUAL_TOOL_CLAMP_PB	manual-clamp-pb	Manual tool clamp switch at head (PLC X01B TCFS.M)	PROPOSED
IN31	SERVO_READY	servo-ready	Servo drives ready relay contact	COMMISSIONING_PENDING

10. 7i84U-A outputs OUT0-OUT15

Channel	Signal ID	HAL net	Field point / load	Status
OUT0	SPINDLE_FWD	spindle-fwd	FR-SX forward input	COMMISSIONING_PENDING
OUT1	SPINDLE_REV	spindle-rev	FR-SX reverse input	COMMISSIONING_PENDING
OUT2	SPINDLE_ENABLE	spindle-run-output	FR-SX enable input	COMMISSIONING_PENDING
OUT3	HYD_PUMP_ON	hyd-pump-on	Hydraulic + head-lube pump contactor (PLC Y096 HYD.M)	PROPOSED
OUT4	SPINDLE_ORIENT_CMD	spindle-orient-cmd	FR-SX orient command (PLC Y093 ORCM1.M)	PROPOSED
OUT5	SPINDLE_ORIENT_LOGEAR	orient-lo-gear	Low-gear orient assist (PLC Y094 CTL.M)	PROPOSED
OUT6	Z_BRAKE_REL_ENABLE	z-brake-release	N1J-L2-201 Z brake release	COMMISSIONING_PENDING
OUT7	GEAR_HI_SOL	gear-hi-sol	RLY-1 to SOL-13 Fujikoshi hydraulic valve	COMMISSIONING_PENDING
OUT8	GEAR_LO_SOL	gear-lo-sol	RLY-2 to SOL-12 Fujikoshi hydraulic valve	HOLD_CONFLICT
OUT9	TOOL_CLAMP_SOL	tool-clamp-sol	RLY-3 to SOL-10 Fujikoshi hydraulic valve	COMMISSIONING_PENDING
OUT10	TOOL_UNCLAMP_SOL	tool-unclamp-sol	RLY-4 to SOL-10 Fujikoshi hydraulic valve	COMMISSIONING_PENDING
OUT11	COOLANT_ON	flood-coolant	Coolant pump relay	COMMISSIONING_PENDING
OUT12	MIST_COOLANT	mist-coolant	Mist coolant relay (was documented as lube pump; verify at cabinet)	HOLD_CONFLICT
OUT13	MAG_CW_SOL	mag-cw-sol	ATC magazine CW rotation relay (SOL-8A per connector_crossref.md, unverified)	HOLD_CONFLICT
OUT14	MAG_CCW_SOL	mag-ccw-sol	ATC magazine CCW rotation relay (SOL-8B per connector_crossref.md, unverified)	HOLD_CONFLICT
OUT15	ALARM_OUT	alarm-out	Alarm light or horn	OPTIONAL_VERIFY

11. 7i84U-B TB1 power map

Channel	Signal ID	HAL net	Field point / load	Status
TB1 pins 3/4 VFIELDA (+24 V field power, 5-28 VDC)	SEVENI84UB_FIELD_A_24V	none	Field power bank A (TB3 I/O)	COMMISSIONING_PENDING
TB1 pins 1/2 VFIELDB (+24 V field power, 5-28 VDC)	SEVENI84UB_FIELD_B_24V	none	Field power bank B (TB2 I/O)	COMMISSIONING_PENDING
TB1 pin 5 VIN (+24 V logic power)	SEVENI84UB_VIN_24V	none	Field I/O logic power	COMMISSIONING_PENDING
TB1 pins 6/7/8 GND (VIN/VFIELD common)	SEVENI84UB_GND	none	VIN/VFIELD common return	COMMISSIONING_PENDING

B-CARD TB1

Pins 1/2 are both VFIELDB positive; pins 3/4 are both VFIELDA positive; pin 5 is VIN; pins 6/7/8 are common. Use 5-28 VDC and record W1.

12. 7i84U-B inputs IN0-IN31

Channel	Signal ID	HAL net	Field point / load	Status
IN0	X_LIMIT_PLUS	limit-x-plus	X positive overtravel limit (NC)	PROPOSED
IN1	X_LIMIT_MINUS	limit-x-minus	X negative overtravel limit (NC)	PROPOSED
IN2	Y_LIMIT_PLUS	limit-y-plus	Y positive overtravel limit (NC)	PROPOSED
IN3	Y_LIMIT_MINUS	limit-y-minus	Y negative overtravel limit (NC)	PROPOSED
IN4	Z_LIMIT_PLUS	limit-z-plus	Z positive overtravel limit (NC)	PROPOSED
IN5	Z_LIMIT_MINUS	limit-z-minus	Z negative overtravel limit (NC)	PROPOSED
IN6	X_HOME	home-x	X home switch (LS-42 assumed)	PROPOSED
IN7	Y_HOME	home-y	Y home switch (LS-52 assumed)	PROPOSED
IN8	Z_HOME	home-z	Z home switch (LS-62 confirmed TB-51)	PROPOSED
IN9	AIR_OK	air-ok	Machine air-pressure OK switch	PROPOSED
IN10	SEVENI84UB_IN10_SPARE	none	Spare input on 7i84U-B	SPARE
IN11	SEVENI84UB_IN11_SPARE	none	Spare input on 7i84U-B	SPARE
IN12	SEVENI84UB_IN12_SPARE	none	Spare input on 7i84U-B	SPARE
IN13	SEVENI84UB_IN13_SPARE	none	Spare input on 7i84U-B	SPARE
IN14	SEVENI84UB_IN14_SPARE	none	Spare input on 7i84U-B	SPARE
IN15	PROBE_SKIP1	probe-in	Renishaw MP-3 probe SKIP1 (PLC X03F SKIP1.M)	PROPOSED
IN16	SEVENI84UB_IN16_SPARE	none	Spare input on 7i84U-B	SPARE
IN17	SEVENI84UB_IN17_SPARE	none	Spare input on 7i84U-B	SPARE
IN18	SEVENI84UB_IN18_SPARE	none	Spare input on 7i84U-B	SPARE
IN19	SEVENI84UB_IN19_SPARE	none	Spare input on 7i84U-B	SPARE
IN20	SEVENI84UB_IN20_SPARE	none	Spare input on 7i84U-B	SPARE
IN21	SEVENI84UB_IN21_SPARE	none	Spare input on 7i84U-B	SPARE
IN22	SEVENI84UB_IN22_SPARE	none	Spare input on 7i84U-B	SPARE
IN23	SEVENI84UB_IN23_SPARE	none	Spare input on 7i84U-B	SPARE
IN24	SEVENI84UB_IN24_SPARE	none	Spare input on 7i84U-B	SPARE
IN25	SEVENI84UB_IN25_SPARE	none	Spare input on 7i84U-B	SPARE
IN26	SEVENI84UB_IN26_SPARE	none	Spare input on 7i84U-B	SPARE
IN27	SEVENI84UB_IN27_SPARE	none	Spare input on 7i84U-B	SPARE
IN28	SEVENI84UB_IN28_SPARE	none	Spare input on 7i84U-B	SPARE
IN29	SEVENI84UB_IN29_SPARE	none	Spare input on 7i84U-B	SPARE
IN30	SEVENI84UB_IN30_SPARE	none	Spare input on 7i84U-B	SPARE
IN31	SEVENI84UB_IN31_SPARE	none	Spare input on 7i84U-B	SPARE

13. 7i84U-B outputs OUT0-OUT15

Channel	Signal ID	HAL net	Field point / load	Status
OUT0	X_DRIVE_ENABLE	x-enable	X servo S-ON to MELDAS DK-427 (X-drive ENA input)	COMMISSIONING_PENDING
OUT1	Y_DRIVE_ENABLE	y-enable	Y servo S-ON to MELDAS DK-427 (Y-drive ENA input)	COMMISSIONING_PENDING
OUT2	Z_DRIVE_ENABLE	z-enable	Z servo S-ON to MELDAS DK-427 (Z-drive ENA input)	COMMISSIONING_PENDING
OUT3	AIR_BLAST	air-blast	SOL-62 via RLY-5 (interposing relay for 100VAC coil)	COMMISSIONING_PENDING
OUT4	TOUCH_SENSOR_BLAST	touch-sensor-blast	SOL-35 via RLY-6 (interposing relay for 100VAC coil)	COMMISSIONING_PENDING
OUT5	TAP_COOLANT_BLAST	tap-coolant-blast	SOL-61 via RLY-7 (interposing relay for 100VAC coil)	COMMISSIONING_PENDING
OUT6	ATC_BARRIER_SOL	atc-barrier	ATC barrier expand solenoid (PLC Y095 TCME.M)	PROPOSED
OUT7	FLOOD_VALVE	flood-valve	Flood coolant valve, separate from pump motor (PLC Y011 FCL)	PROPOSED
OUT8	MAG_COVER_CLOSE_SOL	mag-cover-close	Single energise-to-close magazine cover valve via interposing relay	PROPOSED
OUT9	SEVENI84UB_OUT9_SPARE	none	Spare output on 7i84U-B	SPARE
OUT10	SEVENI84UB_OUT10_SPARE	none	Spare output on 7i84U-B	SPARE
OUT11	SEVENI84UB_OUT11_SPARE	none	Spare output on 7i84U-B	SPARE
OUT12	SEVENI84UB_OUT12_SPARE	none	Spare output on 7i84U-B	SPARE
OUT13	SEVENI84UB_OUT13_SPARE	none	Spare output on 7i84U-B	SPARE
OUT14	SEVENI84UB_OUT14_SPARE	none	Spare output on 7i84U-B	SPARE
OUT15	SEVENI84UB_OUT15_SPARE	none	Spare output on 7i84U-B	SPARE

14. Unassigned, deferred, and P3 rows

Channel	Signal ID	HAL net	Field point / load	Status
none	SPINDLE_ENCODER	none	Machine-side A/B/Z spindle encoder if fitted	UNBOUND
none	CYCLE_START_PB	cycle-start-pb	Operator cycle start pushbutton	DEFERRED
TBD_FROM_IDROM	P3_GPIO_SPARE	none	Spare direct FPGA GPIO on P3 (all pins)	SPARE

15. Conflict and commissioning blocker register

Signal	Card/channel	Field point	Status	Required resolution
SPINDLE_ENCODER	none none	Machine-side A/B/Z spindle encoder if fitted	UNBOUND	Current target leaves P3 empty and requests num_encoders=0. Identify encoder model/electrical format and select a compatible receiver/daughter interface plus IDROM-proven pins before allocation.
GEAR_LO_SOL	7i84U-A OUT8	RLY-2 to SOL-12 Fujikoshi hydraulic valve	HOLD_CONFLICT	Conflict: connector_crossref.md omits SOL-12 and identifies SOL-13 as low; verify both gear-shift coils before wiring
MIST_COOLANT	7i84U-A OUT12	Mist coolant relay (was documented as lube pump; verify at cabinet)	HOLD_CONFLICT	CONFLICT: HAL wires iocontrol.0.coolant-mist to this output; CSV previously carried LUBE_ON. Trace physical load before commissioning; add row to wiring/authority_conflicts.md.
MAG_CW_SOL	7i84U-A OUT13	ATC magazine CW rotation relay (SOL-8A per connector_crossref.md, unverified)	HOLD_CONFLICT	CONFLICT #3 in wiring/authority_conflicts.md: SOL-8A/8B direction mapping unproven. Renamed from atc-fwd to match atc_orient.hal net. Verify direction under controlled commissioning.
MAG_CCW_SOL	7i84U-A OUT14	ATC magazine CCW rotation relay (SOL-8B per connector_crossref.md, unverified)	HOLD_CONFLICT	CONFLICT #3 in wiring/authority_conflicts.md: SOL-8A/8B direction mapping unproven. Renamed from atc-rev to match atc_orient.hal net. Verify direction under controlled commissioning.

16. Evidence-state legend

State	Meaning for wiring/use
PROPOSED	Paper design only; no physical verification
COMMISSIONING_PENDING	Defined but not physically accepted; treated conservatively like proposed
TRACED	De-energized conductor path and device action physically traced
ELECTRICALLY_VERIFIED	Normal/tripped voltage and load behavior measured
HAL_VERIFIED	Physical stimulus and unique HAL response captured
COMMISSIONED	Passed functional and applicable fault-injection acceptance
HOLD_CONFLICT	Contradictory identity/function; do not land or energize
UNBOUND	No hardware terminal/interface assigned
DEFERRED	Outside current scope
SPARE	Unassigned terminal; leave unlabeled